MIL-N-634C
18 November 1977
SUPERSEDING
MIL-N-634B
6 January 1961
(See 6.5)

MILITARY SPECIFICATION

NAVIGATOR'S PLOTTING INSTRUMENT SET AND PLOTTING INSTRUMENTS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers individual plotting instruments and a plotting instrument set fitted within a case and including replacement needle points, leads, and clamp screws used for chart plotting (see 6.1 and 6.2).

2. APPLICABLE DOCUMENTS

2.1 Issue of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
PPP-B-566 - Box, Folding, Paperboard.
PPP-B-585 - Box, Wood, Wirebound.
PPP-B-591 - Box, Fiberboard, Wood-Cleated.
PPP-B-601 - Box, Wood, Cleated-Plywood.
PPP-B-621 - Box, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Shipping, Fiberboard.
PPP-B-676 - Box, Set-Up.
PPP-P-40 - Packaging and Packing of Hand Tools.
PPP-T-60 - Tape, Packaging, Waterproof.
PPP-T-76 - Tape, Packaging, Paper (For Carton Sealing).

MILITARY
MIL-L-10547 - Liners, Case, and Sheet, Overwrap, Water Vaporproof or Waterproof Flexible.

STANDARDS

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
ASTM A276 - Stainless and Heat-Resisting Steel Bars and Shapes, Specification For.
ASTM B121 - Leaded Brass Plates, Sheet Strip, And Rolled Bar, Specification For.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Ship Engineering Center, SEC 6124, Department of the Navy, Washington, D.C. 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6675

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT
Uniform Freight Classification Ratings, Rules and Regulations.

(Application for copies should be addressed to the Uniform Classification Committee Agent, Tarriff Publication Officer, Room 1106, 222 South Riverside Drive, Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), the contractor shall furnish a sample unit for first article inspection and approval (see 4.7 and 6.3).

3.2 Case. The case shall be of the hinged or flap type, and shall not exceed 8 inches long, 3 inches wide, and 1 inch high. The case shall be constructed of wood and shall be covered with a plastic coated fabric. The inside bottom of each case shall be made with recesses shaped so that the instruments and accessories will fit neatly into their respective places with needles in place and at their extreme dimension. The inside of the case, including that part which covers the instruments when the case lid is closed, shall be lined with a dark-colored velvet or velveteen. The case shall be provided with corrosion-resisting steel snaps or fasteners to keep the case closed when not in use.

3.2.1 Contents. The case shall contain a compass, a divider, a screwdriver, two additional leads, a complete additional set of needle points, one of each type and size of knurled clamp screws and a vial for containing the additional leads, needles and clamp screws (see 3.6).

3.3 Compass. Unless otherwise specified (see 6.2.1), the compass shall be made of either brass conforming to ASTM B121, copper alloy No. 342 or 353, 1/2 hard, electroplated with nickel to a minimum thickness of 0.001 inch, or of corrosion-resisting steel conforming to type 304 of ASTM A276. The compass length shall be a minimum of 5-1/2 inches and a maximum of 6-1/2 inches, measured from center of pivot end of legs, exclusive of lead or needle point. The legs shall have a cross section not less than 0.040 square inches and may be one piece or open truss type construction. When truss type construction is used, the material shall be type 304 corrosion resistant steel. The needle point shall be held by a corrosion-resisting steel clamp sleeve, pivoting on the knurled clamp screw affixed to the leg. The needle shall be reversible and replaceable in its carrier. Points shall have an ordinary needle point on one end and a short point with a shoulder on the opposite end to limit the entering depth of the point. The pencil lead shall be held in place in a split clamp sleeve. A knurled clamp screw shall be provided as a means for securing the pencil lead in place. Design and construction shall be such that with lead and needle points extended 1/4 inch from the ends of their respective legs, the compass will close the points to within 1/32 inch of each other.

3.3.1 Compass head. A straightening device, made of corrosion-resisting steel conforming to type 304 of ASTM A276, shall be fitted to keep the head centered and up with reference to the legs of the instrument and provide equal and simultaneous movement of each leg during adjustment. The legs of the compass shall be held together at the pivotal point by a friction joint, the two legs being separated by a steel friction bearing surface attached to one leg. The bearing surfaces shall be parallel and finished to a perfect fit and shall have no rough surfaces, burrs, flash, sharp edges, or tool mark, in order to permit opening and closing with a smooth, uniform, and continuous movement. The type of head construction with only a through bolt as holding and locking device shall not be used.

3.3.1.1 The torque required for opening or closing the compass shall be not less than 40 and not more than 100 ounce-inches both before and after 100 opening and closing cycles. On a single instrument, the range of torque required to open or close the compass from any position shall not exceed 25 ounce-inches both before and after 100 closing cycles. The combined mean of opening and closing torques before 100 opening and closing cycles shall not be changed by more than 20 ounce-inches after 100 opening and closing cycles.

3.4 Divider. The divider shall be identical to the compass (see 3.3), except that both legs shall be fitted to take reversible and replaceable steel needle points, which shall be firmly held in place by knurled clamp screws.

3.5 Needle points, screw studs, and insert lugs. Needle points, screw studs, and insert lugs shall be made of corrosion-resisting steel conforming to type 304 of ASTM A276.

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3.6 Lead and needle point holder. Each set shall contain a tubular holder with a removable cap made of corrosion-resisting metal or plastic. The holder shall contain the additional leads, needle points, and knurled clamp screws specified in 3.2.1.

3.7 Marking.

3.7.1 Instruments. Each compass and divider shall be marked in a plain and permanent manner with the manufacturer's name or identifying trade mark.

3.7.2 Case. The top flap or lid of the case shall have the following notation embossed in gold in the center on the outside surface:

"Navigators Plotting Instrument Set.
National Stock No.
Manufacturer's Name.
Specification or contract No. Year.
U.S."

3.8 Workmanship. Workmanship shall produce instruments of lightness and good balance combined with rigidity in their several parts; joints shall be finely adjusted so as to work freely, smoothly, and rapidly, and retain the parts in any permissible position with proper firmness. There shall be no rough surfaces, burrs, sharp edges, illegibility or tool marks.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 The contractor shall provide and maintain an inspection system in accordance with the data ordering documents included in the contract (see 6.2.2).

4.3 Submission of test reports. Reports covering the inspection and tests for first article inspection and the quality conformance inspections shall be furnished to the procuring activity when specified (see 6.2.2).

4.4 Material inspection. When specified (see 6.2.1), material inspection shall consist of certification (see 6.2.2) supported by verifying data that materials used in the fabrication of the Navigators Plotting Instrument Set and Plotting Instruments are in accordance with the applicable referenced specifications or requirements prior to such fabrication.

4.5 Standard test conditions. All measurements and tests shall be performed under normal room ambient conditions.

4.6 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.7).
(b) Quality conformance inspection (see 4.8).

4.7 First article inspection. First article inspection shall be performed on one complete set, parts, or replacement items, as applicable, and shall consist of the examination and tests specified in 4.9, 4.10, and 4.11.

4.8 Quality conformance inspection. Quality conformance inspection shall consist of the group A and B tests of 4.8.1.2 and 4.8.1.3.

4.8.1 Sampling.

4.8.1.1 Lot. A lot shall consist of complete sets, parts or replacement items, as applicable, offered for delivery at one time and produced on the same facilities, using identical materials, manufacturing and assembly procedures.

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4.8.1.2 Sampling for group A tests. A random sample shall be selected from each lot in accordance with inspection level II of MIL-STD-105. The Acceptable Quality Level (AQL) shall be 1.0 and 4.0 percent defective for major and minor defects, respectively. The samples shall be subjected to the examination and tests of 4.9 and 4.10.

4.8.1.3 Sampling for group B tests. A random sample shall be selected from each lot in accordance with level S-2 of MIL-STD-105. The AQL shall be 2.5 percent defective. The samples shall be subjected to the test of 4.11.

4.8.1.4 Sampling for inspection of preparation for delivery. For purposes of inspection, a completed package shall be considered a unit of product for the packaging examination and a completed pack prepared for shipment shall be considered a unit of product for the packing examination.

4.9 Visual and dimensional examination. The complete sets or, parts and replacement items shall be visually and dimensionally examined to determine defects in accordance with table I.

TABLE I. Classification of defects.

	Defects
Critical	None defined
Major	
101	Design not as specified.
102	Material not as specified. Parts bent, fractured, or not alined.
103	Bearing surfaces not parallel, imperfect fit, rough surfaces burrs, flash, sharp edges, or tool marks.
104	Set screws or thumb screw material not as specified, missing, bent, damaged threads, or binding.
105	Dimensions not within specified tolerance.
106	Finish rough, sharp edge, burrs or tool marks.
107	Instrument marking not as specified, missing or illegible.
108	Case and contents not as specified. Parts missing.
109	Case marking not as specified, missing, not embossed or illegible.
Minor	
201	Package containers and method of box closure not as specified, marking illegible, incorrect, incomplete, or missing.
202	Pack containers not as specified, case liners missing or not as specified, method of box closure, strapping not as specified, gross weight of wood or wood cleated box is in excess of 200 pounds. Marking illegible, incorrect, incomplete, or missing.

4.10 Functional test. The compass and divider shall be subjected to 25 cycles as specified in 4.10(a) and the following characteristics shall be recorded upon completion of test:

(a) Head centered and up with respect to legs of instrument; equal and simultaneous movement of legs during adjustment.
(b) Smoothness of operation.
(c) Visible evidence of abrasion or wear.
(d) Ability to maintain alinement of points within 1/32 inch.
(e) Ability to maintain legs in any position with no loosening of parts.

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4.11 Torque test. The compass and divider shall be tested as follows:

(a) With the compass or divider in a closed position, adjust needle points or pencil lead, as applicable, so that the points just touch. Then with a continuous movement, extend the legs to full open position and then back to the fully closed position. (This constitutes one cycle.)
(b) The divider or compass shall be opened to a point separation of 1/2 inch. One leg shall be held securely while a measured force is applied at right angles to the other leg at a known distance from the pivot point such that the divider or compass is slowly forced open to a point separation of at least 5 inches. The forces required during this opening procedure shall be recorded at the beginning of motion and at equal intervals of point separation (minimum of 5 readings) and converted to torques (ounce-inch) by the distance in inches between the measured force and the pivot point. The force shall then be reversed so that the compass or divider slowly closes to 1/2 inch point separation. The torque shall be established again at equal intervals of point separation (minimum of 5 readings).
(c) Repeat step (a) 100 times.
(d) Repeat step (b).
(e) Calculate average of opening and closing torques established in steps (b) and (d). Subtract the smaller average from the larger.
(f) Any variation from 3.3.1.1 shall be considered a defect.

4.12 Inspection of preparation for delivery. Sample packages and packs and the inspection of preservation, packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements listed in section 2, see 6.4.)

5.1 Packaging and packing.

5.1.1 Packaging. Packaging shall be level A or C as specified (see 6.2.1).

5.1.1.1 Level A. Each set complete with instruments and accessories shall be packed in a snug fitting folding or set-up paperboard box conforming to PPP-B-566 and PPP-B-676 respectively at the option of the supplier. Box closure shall be as specified in the applicable box specification or appendix thereto.

5.1.1.2 Level C. Each set complete with instruments and accessories shall be packaged in accordance with the suppliers' commercial practice.

5.1.2 Packing. Packing shall be level A, B or C as specified (see 6.2.1).

5.1.2.1 Level A. Material, packaged as specified in 5.1.1, shall be packed in any of the following boxes at the option of the contractor.

Specification	Type or class
PPP-B-585	Class 3 use
PPP-B-591	Overseas type
PPP-B-601	Overseas type
PPP-B-621	Class 2
PPP-B-636	Class 2

Shipping containers shall be provided with waterproof case liners conforming to MIL-L-10547. Case liners shall be closed and sealed in accordance with the appendix to MIL-L-10547. Case liners for fiberboard boxes conforming to PPP-B-636 may be omitted provided all corner and edge seams and manufacturer's joints are sealed with minimum 1-1/2 inch wide tape conforming to PPP-T-60 or PPP-T-76. Box closure and strapping shall be as specified in the applicable box specification or appendix thereto. Strapping shall be zinc coated. The gross weight of wood or wood cleated boxes shall not exceed 200 pounds.

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5.1.2.2 Level B. Material, packaged as specified in 5.1.1, shall be packed in any of the following containers at the option of the contractor:

Specification	Type or class
PPP-B-585	Class 1 or 2 use
PPP-B-591	Domestic type
PPP-B-601	Domestic type
PPP-B-621	Class 1
PPP-B-636	Class 1

Box closure shall be as specified in the applicable box specification or appendix thereto. The gross weight of wood or wood-cleated containers shall not exceed 200 pounds.

5.1.2.3 Level C. Material, packaged as specified in 5.1.1, shall be packed in a manner which will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply to the Uniform Freight Classification Rules or other carrier regulations as applicable to the mode of transportation.

5.1.3 Instruments and accessories. Instruments and accessories, when procured separately, shall be preserved and packaged by level A or C; packed by level A, B or C as specified (see 6.2.1).

5.1.4 Marking. In addition to any special marking required by the contract, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Individual replacement instruments parts such as needle points, clamp screws and leads may be ordered separately under this specification as repair parts.

6.2 Ordering data.

6.2.1 Procurement requirements. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Whether plotting instrument set or individual plotting instruments are required (see 1.1 and 6.1).
(c) When samples for first article inspection are required (see 3.1).
(d) When material required for compass is other than as specified (see 3.3).
(e) That the contractor shall certify in writing that the product meets the material requirements of the applicable referenced specifications (see 4.4).
(f) Level of packaging and level of packing required (see 5.1.1, 5.1.2, and 5.1.3).

6.2.2 Data requirements. When this specification is used in a procurement which invokes the provision of the "Requirements for Data" of the Armed Services Procurement Regulations (ASPR), the data identified below, which are required to be developed by the contractor, as specified on an approved Data Item Description (DD Form 1664), and which are required to be delivered to the Government, should be selected and specified on the approved Contract Data Requirement List (DD Form 1423) and incorporated in the contract. When the provisions of the "Requirements for Data" of the ASPR are not invoked in a procurement, the data required to be developed by the contractor and required to be delivered to the Government should be selected from the list below and specified in the contract.

Paragraph	Data requirement	Applicable DID
4.2	Inspection system	UDI-T-23741
4.3	First article and quality conformance inspection reports	UDI-T-23473
4.4	Material certification	UDI-T-23741

(Copies of data item descriptions required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.)

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6.2.3 The data requirements of MIL-N-634 and any task in sections 3,4, or 5 of the specification required to be performed to meet a data requirement may be waived by the procuring/purchasing activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item procured to this specification. This does not apply to specific data which may be required for each procurement regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.4 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.5 Changes from previous issue. The symbol "#" is not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - MD
Navy - SH
Air Force - 99

Preparing activity:
Navy - SH

(Project 6675-0289)

Review activity:
DGSC - GS

User activity:
Navy - AS

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

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